

Data Vortex — Round 1

Exploratory Data Analysis (EDA) Report

Dataset: Social Engine (recovered from node_07)

Files: Social_Engine_Users.csv (1,500 rows), Social_Engine_Posts_Corrupted.csv (12,360 rows)

1. Data Recovery

The raw CSVs were not distributed as files — they were embedded as plain string constants inside the recovery terminal's compiled JavaScript bundle. They were extracted directly from that bundle and stored, unmodified, in **data/raw/**.

2. Data Quality Issues Found & Fixes Applied

Users.csv needed no cleaning — zero missing values, zero duplicate rows, zero duplicate user_id values, no negative follower_count values, and every date parsed cleanly.

Posts.csv had five distinct issues:

Issue	Rows affected	Fix applied
Exact duplicate rows	360	Dropped — confirmed identical copies, not conflicting records
& entity in text_content	328	HTML-unescaped to & — only entity type present
Timestamps in 3 formats (Unix epoch / ISO 8601 / DD-MM-YYYY)	12,000 (post-dedup)	Parsed into one consistent datetime column; date-only rows flagged via timestamp_time_estimated
Negative likes values	509	Converted to absolute value — magnitudes matched the valid-likes distribution (sign-flip corruption)
Missing platform / text_content / likes (~15% each)	1,784 / 1,688 / 1,814	Left as NULL — missingness verified statistically independent across columns; no basis to impute

A referential-integrity check confirmed every user_id in Posts.csv exists in Users.csv — zero orphan rows. Full reproducible logic lives in scripts/clean.py.

3. Platform Distribution

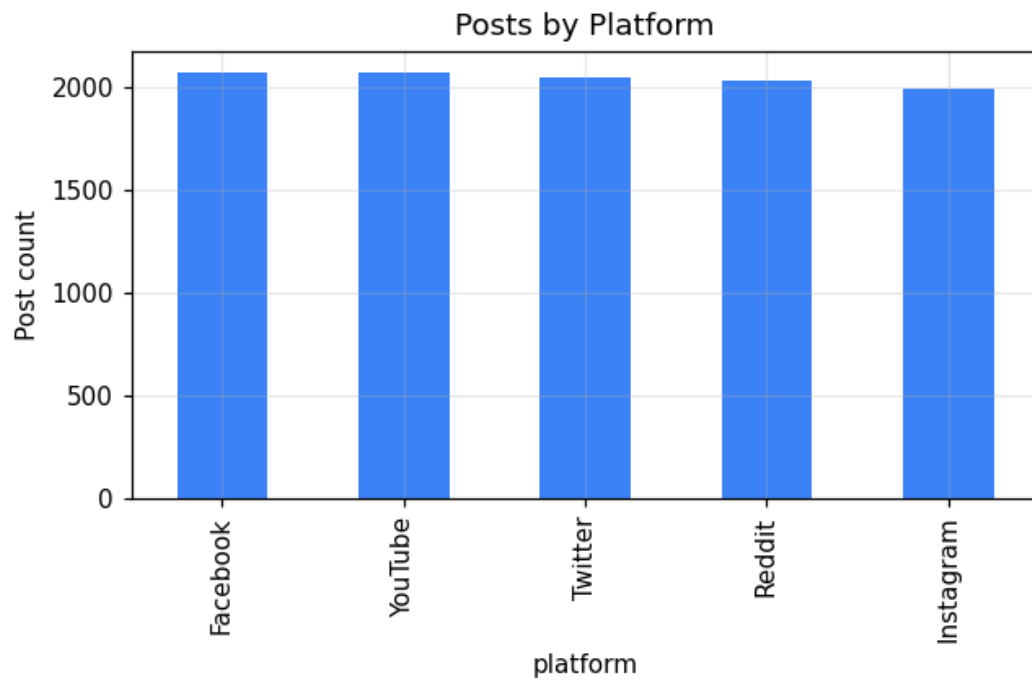


Figure 1 — Post volume is almost perfectly even across all five platforms (coefficient of variation under 2%). No dominant platform.

4. Posting Volume Over Time

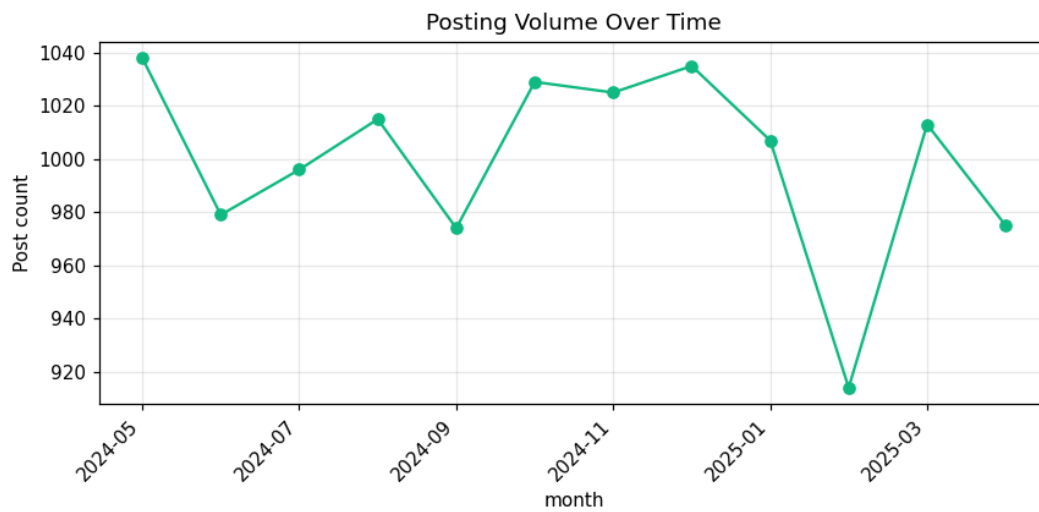


Figure 2 — Monthly post counts fluctuate within a narrow band with no clear seasonal trend or growth/decline pattern.

5. Engagement Metrics: Likes, Shares, Comments

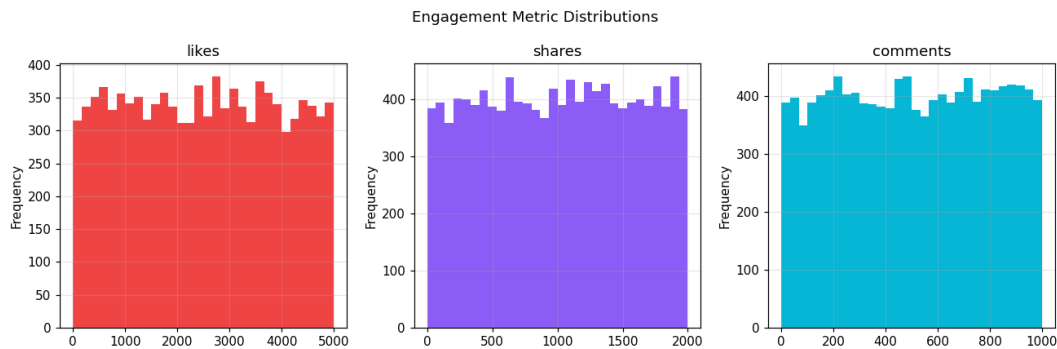


Figure 3 — All three engagement metrics are uniformly distributed over their respective ranges, each with no long tail.

Real organic engagement data is almost always long-tailed (most posts get little engagement, a few go viral) — a flat/uniform shape is the signature of independently randomly generated values rather than genuine user behavior.

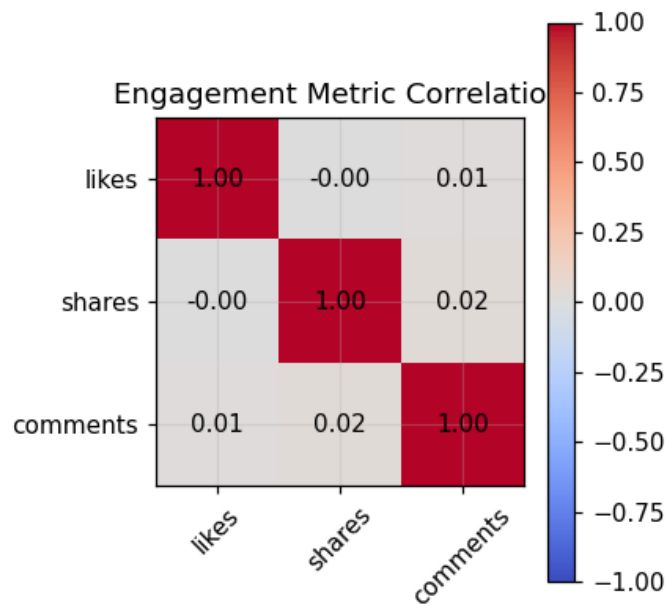


Figure 4 — Pairwise correlation between likes, shares, and comments is effectively zero (all $|r| < 0.02$).

In real engagement data these three metrics normally move together (a post with many likes tends to also get more shares and comments). Their near-total independence here reinforces that these fields were generated independently at random rather than reflecting organic post performance.

6. Follower Counts

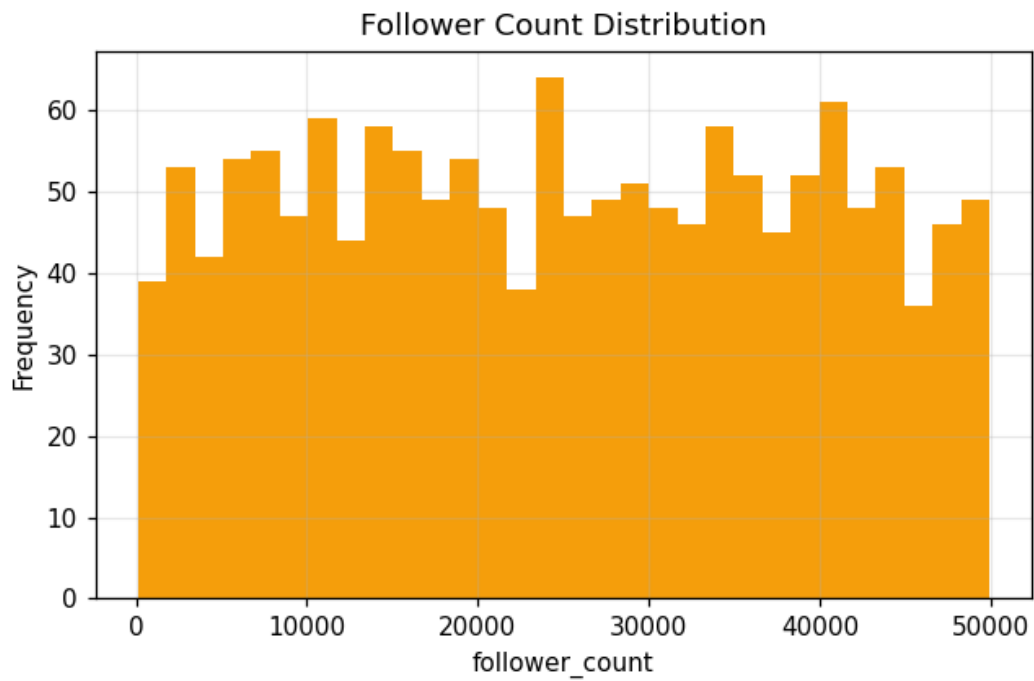


Figure 5 — follower_count is close to uniform across its range (109–49,944), with mean and median nearly equal.

There's no "mega-influencer" long tail — a real social platform's follower distribution is typically extremely skewed. Its absence here is consistent with synthetic/randomly generated data.

7. User Locations & Languages

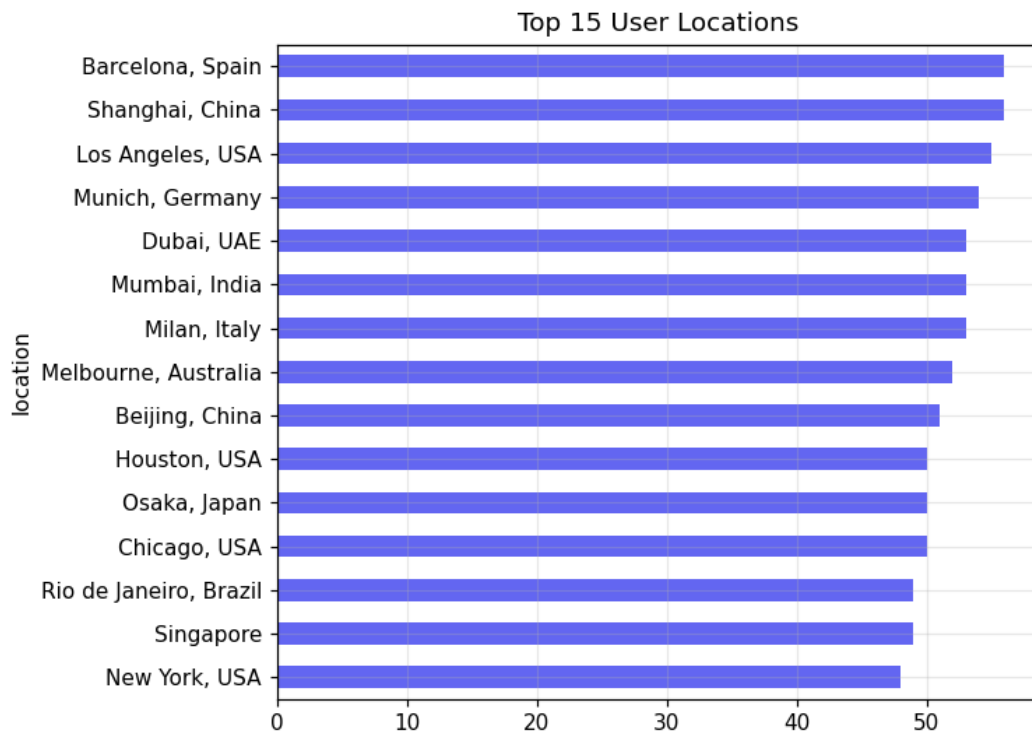


Figure 6 — Top 15 user locations by city.

Aggregating city-level location values up to country level: **USA is the single largest user location** (203 users), followed by China (107), Germany (97), Italy (96), and Japan (94), with a broad spread across roughly a dozen more countries — no single non-US country comes close to USA's share, but the rest of the world is fairly evenly distributed rather than concentrated in one or two other places.

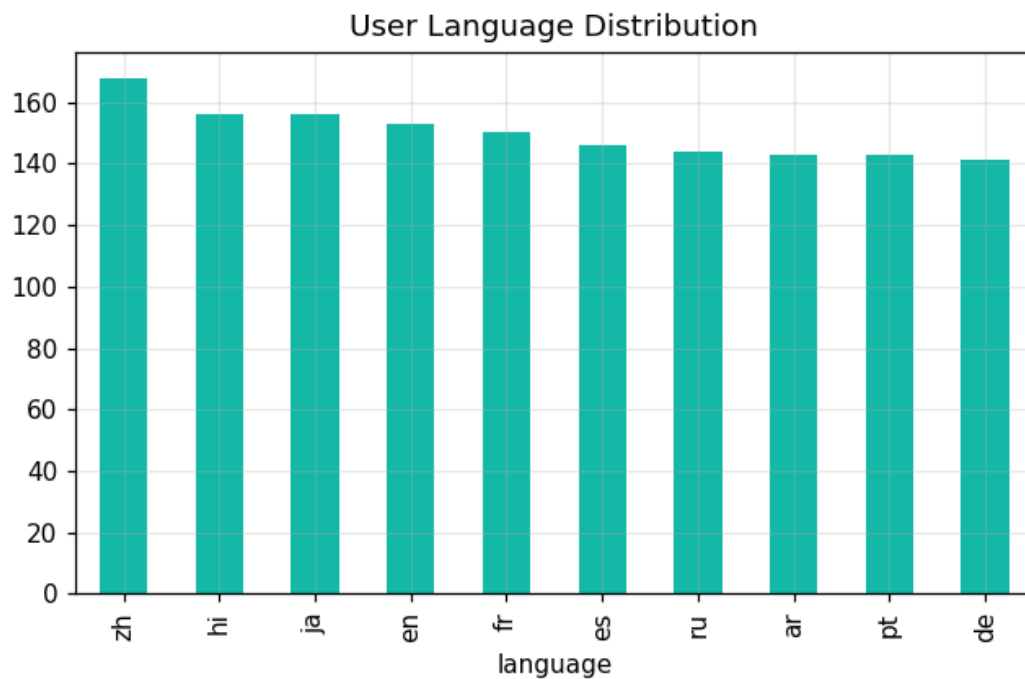


Figure 7 — User language distribution.

8. Missingness Pattern

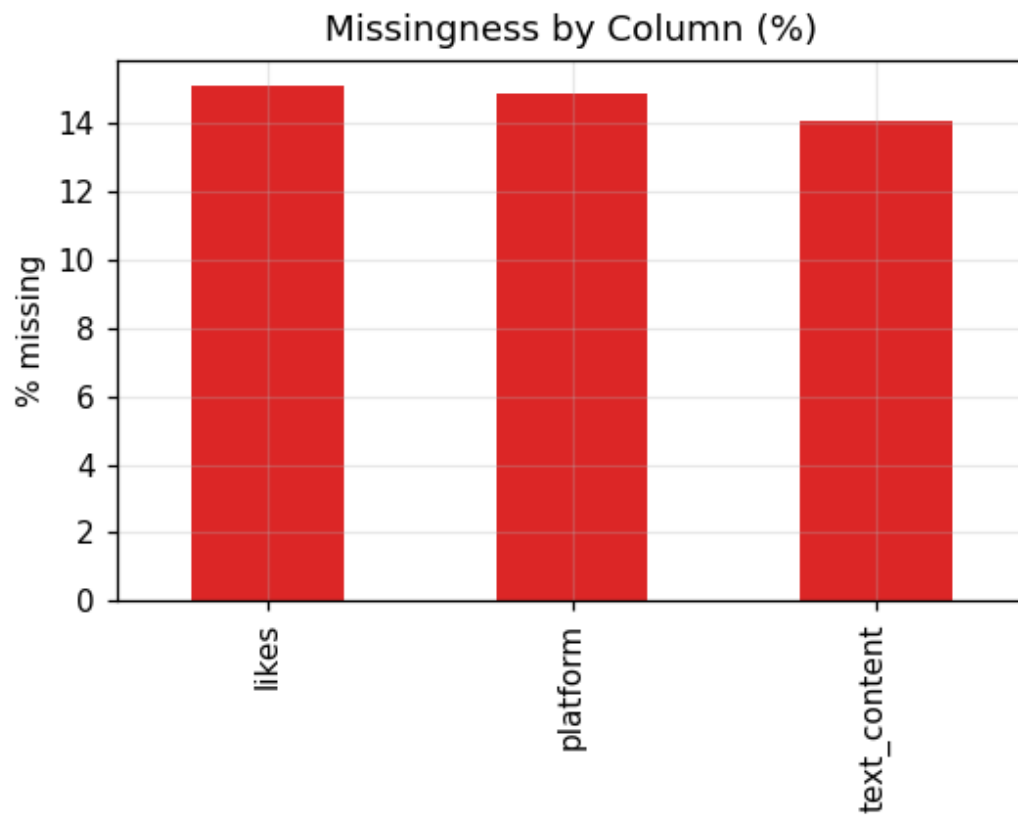


Figure 8 — `platform`, `text_content`, and `likes` are each missing in roughly 14–15% of rows.

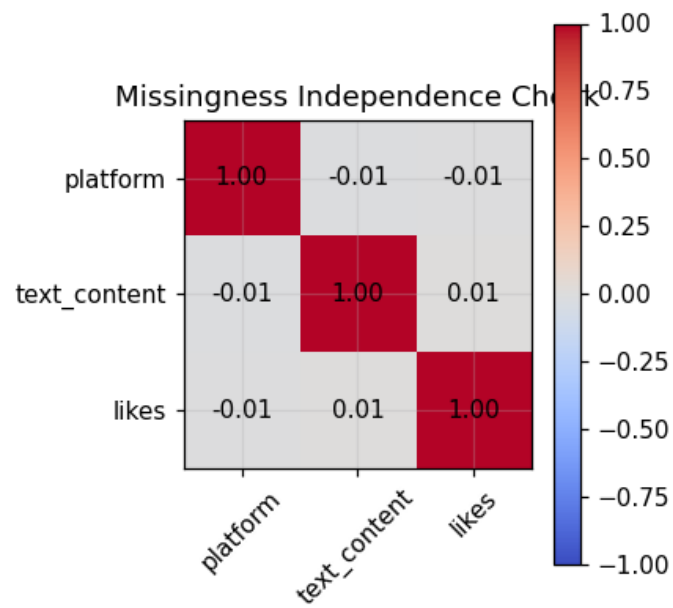


Figure 9 — Pairwise correlation between missingness indicators is essentially zero (all $|r| < 0.02$).

This confirms the missingness is statistically independent across the three fields — not a correlated block that would suggest one shared root cause (e.g. a batch of posts that failed to log entirely). That's why each was left as NULL rather than imputed: there's no defensible signal to guess a value from, and fabricating one would misrepresent the data.

9. Summary of Key Findings

- **Posts.csv shows signs of synthetic generation** — engagement metrics and follower_count are all uniformly distributed with near-zero pairwise correlation, the opposite of the long-tailed, correlated pattern real social-engagement data usually shows.
- **Platform and posting volume are both evenly distributed** — no dominant platform, no seasonal or growth trend.
- **USA is the single largest user location**; the remainder of users are spread broadly across ~15 other countries with no other single country dominating.
- **All corruption in Posts.csv was mechanical and reversible** — duplicate rows, one HTML entity type, three parseable timestamp formats, and a sign-flip on likes — none required guessing or fabricating data to fix.
- **Missing values were left as NULL** because their missingness is independent across columns and carries no signal to impute from.

All numeric detail behind every chart is available in reports/eda_summary.json, generated by scripts/eda.py.